

SEDAR 14: Caribbean Yellowfin Grouper, Mutton Snapper, and Queen Conch

Caribbean Yellowfin Grouper

Data Workshop

Life History

The life history subgroup made several research recommendations pertaining to yellowfin grouper. These are prioritized below.

Early life History

- 1) Conduct studies on temporal (intra- and inter-annual) variability of oceanographic processes in relation to larval dispersal to quantify the degree of connectivity between platforms of the currently managed stock units.
- 2) Examine early larval dispersal patterns (post fertilization to pre-flexion) using genetic markers.
- 3) Identify essential habitats according to life history stage, including critical recruitment and post-settlement (nursery) habitats.

Adult Populations

- 4) Identify additional past and present spawning aggregation sites and characterize migration corridors.
- 5) Define the spatial scale of migrations by individuals participating in spawning aggregations through tag and release studies.
- 6) Evaluate the potential to use visual census data obtained from spawning aggregations as fisheries independent data for assessing stock status (i.e. sex ratio, average size, density) and for monitoring populations.

Stock Identification

- 7) Investigate population genetic structure of yellowfin grouper “stocks” within the US Caribbean and in relation to the wider Caribbean.
- 8) Examine ontogenetic shifts in habitat usage and diel foraging patterns

Commercial

1. Continuous biological sampling in the Virgin Islands at sufficient levels to adequately characterize size and age composition.

2. Link biostatistical data for a fishing trip from Puerto Rico to all of the landings records for that trip.
3. Ensure that the catch and effort data of individual fishers in Puerto Rico can be identified over time.
4. Eliminate the need for expansion factors by obtaining information on all landings.

Recreational

1. Conduct surveys to estimate the magnitude of the U. S. Virgin Islands recreational landings for all species including conch and lobster. It is possible that using a Virgin Islands contractor would improve the likelihood of success of the survey.
2. Include conch and lobster in the MRFSS for Puerto Rico.
3. To adequately characterize catch rates and sizes of mutton snapper caught by recreational anglers in Puerto Rico, very substantial increases in dockside sampling will be needed.

Indicators of Population Abundance

1. Fisheries-independent survey efforts currently rarely include stations in deep water, the preferred habitat of adult mutton snapper and adult yellowfin grouper. In addition, large aggregations of queen conch have been reported in deep water by commercial fishers. The group highly recommends the initiation and continued funding of such surveys. As trends can be regional in nature, the group highly recommends that such surveys be conducted throughout Puerto Rico and the US Virgin Islands.
2. The commercial landings data from Puerto Rico and the US Virgin Islands have been incompletely entered and a variety of problems are known to exist in those data. The group strongly recommends that every effort be made to resolve the problems with those data. This should include extensive meetings with port samplers and others familiar with the US Caribbean fisheries.
3. The group recommends that tag-recapture studies of mutton snapper, yellowfin grouper, and queen conch be conducted in Puerto Rico and the US Virgin Islands to determine habitat utilization and movement of those species.
4. Ongoing long-term monitoring studies should be expanded spatially and include data useful for stock assessment, e.g. size-frequency and density information.
5. It is suggested that areas exploited by fishermen be compared to those areas where monitoring has been ongoing to further knowledge of essential habitat for these species and improve the design of monitoring efforts (i.e., ensure that monitoring is reflective of fished conditions).

6. The group recommends that efforts be made to monitor spawning aggregations of finfish to improve measures of population abundance. Collection of historical indicators of spawner abundance (e.g., directed visual census, analysis of catch statistics for spawning peaks, etc).
7. The group encourages the collection and documentation, for this and future Caribbean assessments, of historical information for qualitative and/or quantitative comparisons of current conditions.

Assessment Workshop

The AWP recommends collecting species level information on commercial and recreational harvest in the US Virgin Islands.

The AWP recommends collecting biological samples to characterize commercial and recreational catches in the US Virgin Islands and Puerto Rico.

The AWP recommends continuation of the survey efforts directed at the Grammanik Bank spawning aggregation as a potential source of yellowfin grouper trends that reflect a potentially important population component.

The AWP recommends developing specific surveys to evaluate species such as yellowfin grouper which rarely occur in general surveys but are known to seasonally aggregate.

The AWP recommends developing research and monitoring programs that enable quantitatively evaluating management actions such as seasonal and area closures, especially as such actions can significantly alter fishery operations and limit traditional data collection approaches.

The AWP recommends pursuing alternative assessment methods for evaluating the status of stocks such as yellowfin grouper that are not commonly encountered by either fishery-dependent or fishery-independent sampling and monitoring programs.

The AWP recommends devoting effort to characterizing basic catch, biological, and survey data availability before recommending SEDAR assessments of stocks that have never been quantitatively assessed. Such work should be considered between scheduled SEDAR assessment projects or perhaps in lieu of a project dedicated to a particular species.

The AWP recommends a complete review of the potential data collection programs, including commercial and recreational catch, biostatistical sampling and fishery-independent surveys for Puerto Rico and US Virgin Islands with the purpose of identifying what relevant information could be obtained and modifying sampling procedures accordingly, including the identification of key economic and ecological indicator species.

The efforts to analyze the available data were greatly enhanced by the presence of local fishers and agency representatives. However, there was no local representative from the USVI Division

of Fish and Wildlife assigned to the AWP and the sole Puerto Rico representative could not attend the full term of the meeting. There must be greater buy-in from the local agencies such that knowledgeable representatives are present for the full term of the meeting. Furthermore, greater efforts should be made to attract and secure participation of local fishers.

Review Workshop

The Review panel replied to Term of Reference 9.

9. Consider the research recommendations provided by the Data and Assessment workshops and make any additional recommendations warranted. Clearly indicate the research and monitoring needs that may appreciably improve the reliability of future assessments. Recommend an appropriate interval for the next assessment and indicate whether a benchmark or update assessment should be considered.

The Review Panel agrees with the points put forward by the Data Workshop and Assessment Workshop.

In addition it was the opinion of the RP that:

1. Tagging data should also be considered in relation to obtaining information on growth rates of yellowfin grouper. In addition to be of general life history interest this will also be of importance in relation to validation of otolith age determination.

Because yellowfin grouper is a quite rare species it might be considered to use some kind of Data Storage Tags in order to obtain as much information from each individual fish tagged as possible. If sufficient number of yellowfin grouper can be caught and tagged (with ordinary tags), annual tagging programs to reveal stock size and fishing mortality could be considered for yellowfin grouper. Because yellowfin grouper is quite rare to catch it might be practical to tag several species (with similar lack of life history knowledge for which tagging studies are potential appropriate) at the same time.

2. An internet setup could be explored, where anglers and maybe divers report their catches of yellowfin grouper (and other relevant species) as well as additional information directly on forms on the internet. Such internet systems are used with success in other places in the world to report fish catches, especially for large and rare species like yellowfin grouper. Such an approach should be accompanied with various test and checks to estimate reporting rates by segments of anglers and divers, etc., so that total catches from relevant segments of the fishers can be estimated in a proper way. The internet is also an effective tool for communicating with the data suppliers, for instance about how to report, the results of the reports and ongoing tagging experiments.

Of all these recommendations including those of Data Workshop and Assessment Workshop, the Panel regards the following to be of the highest priority:

- the improvements of sampling from the fishery (both commercial and recreational) including biological measurement;
- tagging studies to reveal stock structure, population size, annual fishing mortality and life history parameters;
- improving fisheries-independent surveys;
- resolving the problems in the commercial landings data base.

Recommendations for Future SEDAR Assessments

The Review Panel recommends that the assessment and management of inshore and reef fish in the Caribbean should follow a multi-species, mixed fishery approach appropriate to the conditions of coastal tropical fisheries. It is therefore recommended that the scope and timing of the next Assessment Workshop is established following an inter-sessional workshop within the next 12 - 18 months to evaluate the information available to support such an approach. Specifically, the workshop should identify the relative abundance, potential vulnerability to exploitation and type and quality of data available for each species, potential indicator species for which it may be possible to provide reliable single-species assessments and benchmarks, and procedures and data-needs for deriving indicators and benchmarks at the fish community level.

Mutton Snapper

Data Workshop

Early Life History

1. Conduct studies on temporal (intra- and inter-annual) variability of oceanographic processes in relation to larval dispersal to quantify the degree of connectivity between platforms of the currently managed stock units.
2. Examine early larval dispersal patterns (post fertilization to pre-flexion) using genetic markers and otolith microchemistry where possible.
3. Identify essential habitats according to life history stage, including critical recruitment and post-settlement (nursery) habitats.

Adult Populations

4. Identify additional past and present spawning aggregation sites and characterize migration corridors.

5. Define the spatial scale of migrations by individuals participating in spawning aggregations through tag and release studies.
6. Evaluate the potential to use census data obtained from spawning aggregations as fisheries independent data for assessing stock status (i.e. sex ratio, average size, density) and for monitoring populations.
7. Investigate population genetic structure of mutton snapper “stocks” within the U.S. Caribbean and in relation to the wider Caribbean.
8. Examine ontogenetic shifts in habitat usage and diel foraging patterns.

Commercial

5. Continuous biological sampling in the Virgin Islands at sufficient levels to adequately characterize size and age composition.
6. Link biostatistical data for a fishing trip from Puerto Rico to all of the landings records for that trip.
7. Ensure that the catch and effort data of individual fishers in Puerto Rico can be identified over time.
8. Eliminate the need for expansion factors by obtaining information on all landings.

Recreational

4. Conduct surveys to estimate the magnitude of the U. S. Virgin Islands recreational landings for all species including conch and lobster. It is possible that using a Virgin Islands contractor would improve the likelihood of success of the survey.
5. Include conch and lobster in the MRFSS for Puerto Rico.
6. To adequately characterize catch rates and sizes of mutton snapper caught by recreational anglers in Puerto Rico, very substantial increases in dockside sampling will be needed.

Indicators of Population Abundance

1. Fisheries-independent survey efforts currently rarely include stations in deep water, the preferred habitat of adult mutton snapper and adult yellowfin grouper. In addition, large aggregations of queen conch have been reported in deep water by commercial fishers. The group highly recommends the initiation and continued funding of such surveys. As trends can be regional in nature, the group highly recommends that such surveys be conducted throughout Puerto Rico and the US Virgin Islands.
2. The commercial landings data from Puerto Rico and the US Virgin Islands have been incompletely entered and a variety of problems are known to exist in those data. The group strongly recommends that every effort be made to resolve the problems with those

data. This should include extensive meetings with port samplers and others familiar with the US Caribbean fisheries.

3. The group recommends that tag-recapture studies of mutton snapper, yellowfin grouper, and queen conch be conducted in Puerto Rico and the US Virgin Islands to determine habitat utilization and movement of those species.
4. Ongoing long-term monitoring studies should be expanded spatially and include data useful for stock assessment, e.g. size-frequency and density information.
5. It is suggested that areas exploited by fishermen be compared to those areas where monitoring has been ongoing to further knowledge of essential habitat for these species and improve the design of monitoring efforts (i.e., ensure that monitoring is reflective of fished conditions).
6. The group recommends that efforts be made to monitor spawning aggregations of finfish to improve measures of population abundance. Collection of historical indicators of spawner abundance (e.g., directed visual census, analysis of catch statistics for spawning peaks, etc).
7. The group encourages the collection and documentation, for this and future Caribbean assessments, of historical information for qualitative and/or quantitative comparisons of current conditions.

Assessment Workshop

Table 12 provides a comprehensive overview of the availability of information for U.S. Caribbean mutton snapper populations. This table in addition to the following discussion provides a synthesis of the groups thoughts regarding sufficiency and quality of the data available for use in evaluating the stock status of the mutton snapper population in this region. Due to the current categorization of mutton snapper as undergoing overfishing, this species should be prioritized in all data collection efforts in the US Caribbean both in dependent and fishery independent programs. Obtaining information required to assess the impact of regulations on management measures is needed. Targeted research efforts are needed to determine relative abundance, CPUE, length and age structure of catch for all commercial and recreational gears used to harvest mutton snapper. The group noted the need to monitor population densities at seasonal closed areas to open areas to determine effects of management and to monitor compliance. The only area closure for mutton snapper is off St. Croix and the closure has been in place since 1993. There has been no monitoring in this area since the closure took effect. In addition there is no current mechanism of enforcing the spawning seasonal closure.

Review Workshop

The Review panel replied to Term of Reference 9.

9. Review the research recommendations provided by the Data and Assessment workshops and make any additional recommendations warranted. Clearly indicate the research and monitoring needs that may appreciably improve the reliability of future assessments. Recommend an appropriate interval for the next assessment.

The RP reviewed the wide range of research recommendations provided by the DW and AW in relation to immediate and longer-term needs for improving the assessment of the stocks and the provision of management advice. The RP provided additional recommendations where appropriate. The research recommendations are reviewed in a separate section of this report.

The RP recommends that the assessment and management of inshore and reef fish in the Caribbean should follow a multi-species, mixed fishery approach appropriate to the conditions of coastal tropical fisheries. It is therefore recommended that the scope and timing of the next AW is established following an intersessional workshop within the next 12 – 18 months to evaluate the information available to support such an approach. Specifically, the workshop should identify the relative abundance, potential vulnerability to exploitation and type and quality of data available for each species, potential indicator species for which it may be possible to provide reliable single-species assessments and benchmarks, and procedures

Review Panel research recommendations

The DW and AW reports provided a wide range of research recommendations related to biology, fishery data, fishery-independent data and assessment methods for mutton snapper. The recommendations were scattered throughout the reports, but without any prioritization according to short-term and longer-term needs or any indication of the extent to which the results could improve the assessment and management of the stocks. The RP recommends that future DW and AW reports provide a single section collating all recommendations, with priorities and expected contribution of the results clearly identified.

The following sections give the combined DW and AW recommendations for different research areas. In each case these are followed by RP evaluations and consolidated recommendations for data collection and research that is needed to address the deficiencies in data and understanding that are impeding the evaluation of stock status and development of appropriate management measures. In some cases similar recommendations appear in different guises in different parts of the DW and AW reports and the RP has taken the liberty of merging and rewording these as appropriate, and summarizing some of the other recommendations.

DW & AW Workshop recommendations on fishery-dependent data

- Biological sampling at USVI to characterize size and age composition.
- Ensure that the catch and effort data of individual fishers in Puerto Rico can be identified over time.
- Eliminate the need for expansion factors by obtaining information on all landings; resolve other problems with data through extensive meetings with port samplers and others familiar with US Caribbean fisheries.

- Targeted research efforts to determine relative abundance, CPUE, length and age structure of catch for all commercial and recreational gears;
- Collection of species landings data at resolution to allow CPUE data for each gear; need to identify each individual fisher, location/date of catch, and depth where possible.
- Estimate CPUE in terms of numbers and biomass; estimate effort as hook-hours and trap soak times;
- Where appropriate, collection of discards data and fate (dead or alive) of discards;
- Review of field methods and protocols for fishery data collection throughout

Caribbean;

- Review catch sampling intensity protocols;
- Evaluate impacts of management measures, particularly closed areas

The RP considers the improvement in the accuracy and coverage of fishery data to be of very high priority for the fisheries of Puerto Rico and the U.S. Virgin Islands, and endorses the DW and AW recommendations. The RP makes the following consolidated recommendations:

- i) Ensure accurate recording of data by species in all areas.
- ii) Development of a random fishery sampling scheme, stratified by appropriate areas/gears/seasons, to provide valid statistical estimates of catches and size compositions by species, and fishing effort, with high spatial and temporal resolution.
- iii) Continued improvement of log-book reporting schemes and improvements in methods for expanding reported landings to the total fishery, for example by stratifying by port.
- iv) Evaluation of the representativeness of the reported fishery data, for example by interviewing fishermen who have submitted log sheets in recent years but did not before.
- v) Identification of fishing effort units (e.g. soak time for traps; hook-hours) that are most likely to provide a linear relationship between CPUE and population abundance, and the capturing of historical TIP data on landing weight per trip for trips with soak time or other effort data
- vi) Collection of covariates (e.g. depth) to help explain variability in CPUE data
- vii) Accurate documentation of changes over time in fishing effort, fishing gears and their deployment, species targeting and fish-location technology (e.g. GPS), to help interpret CPUE data and identify periods when catchability may have changed.
- viii) The Panel agrees that standardized sampling protocols and systems for Quality Assurance / Quality Control of data are needed for data collection throughout the Caribbean.
- ix) Involvement of fishers in data collection schemes, including investigating the potential for web-based systems for capturing fisher's data and other information.

DW & AW Workshop recommendations on the recreational fishery

- Conduct surveys to estimate magnitude of USVI recreational landings for all species (use a USVI contractor) (To adequately characterize catch rates and sizes of mutton snapper caught by recreational anglers in Puerto Rico, very substantial increases in dockside sampling will be required.)

- MRFSS program should add additional survey attributes to draw out information on mutton snapper throughout US Caribbean; increase MRFSS intercepts to improve sample sizes. The RP endorses recommendations to collect relevant data on recreational fishing. Data on recreational fishery catches of mutton snapper are limited to the recent period of the MRFSS survey (2000 onwards for Puerto Rico, 2000 only for USVI). Although the precision of estimates of fish catches is quite low (CV's = 30-50%), recreational fishing appears to be an important source of mortality (6,000 – 25,000 fish killed per year off Puerto Rico), and shore fishermen appear to target mainly juvenile mutton snapper. Improvements in the coverage and intensity of the Puerto Rico sampling scheme and restarting the USVI scheme would contribute significantly to the accuracy of removals estimates from the stocks. Shore-angling catch rates may indicate recruitment trends. As with the commercial fishery, involvement of the angling community in data collection schemes would be beneficial, potentially making use of web-based systems.

DW & AW Workshop recommendations on fishery independent data

- Initiate surveys in deeper water, the preferred habitat of adult mutton snapper.
- Identify essential habitats according to life history stage, including critical recruitment and post-settlement (nursery) habitats.
- Monitor spawning aggregations for density (abundance indices), and collection of population parameters such as sex ratio and size of fish.
- Collection and documentation of historical information for qualitative and/or quantitative comparisons of current conditions; collation of historical indicators of spawner abundance
- Continue and enhance fishery independent programs including spawning aggregations and collection of data on size of individuals, depth, time of day, habitat; use of visual counts or directed gear sampling;

The RP encourages the development of fishery independent surveys using fishing gears or direct observation, provided the surveys adequately cover the range of the target species and are capable of providing abundance indices or raised abundance estimates with acceptable accuracy. The RP recognizes that such surveys require substantial investment to achieve the necessary spatial coverage, and will benefit from existing studies and fisher's knowledge to identify strata for visual or fishing surveys of spawning fish.

The DW listed 14 different sources of fishery independent data from different areas around Puerto Rico and the U.S. Virgin Islands, but only five appear to provide data on mutton snapper, mainly in the U.S. Virgin Islands. In general the surveys tend to be localized and observations of mutton snapper can be low. Diver surveys using volunteer divers on the REEF program indicate (other than in 2006) an increase in abundance of mutton snappers at inshore sites off the U.S. Virgin Islands, showing a similar general pattern to the Puerto Rico commercial trap fishery CPUE. The existing surveys should be reviewed to establish areas that could be targeted for

systematic dive surveys, and to determine the survey effort required to achieve specified precision levels. Discussions at the Review meeting indicated that surveys at times of year when the fish are more dispersed may provide more precise abundance indices than surveys of spawning aggregations. Occupancy of spawning sites will also be strongly affected by spawning behavior and the environmental triggers for spawning.

On the other hand, surveys designed to collect data on parameters such as relative size composition of mature fish, may benefit from taking place on known spawning sites at spawning time. The design of surveys therefore needs to be linked clearly to their objectives.

The RP recommends investigation of other methods for fishery-independent stock monitoring, for example beach-seine surveys to provide recruitment indices for mutton snapper and other species and tag-release programs to estimate mortality rates as well as fish movements. Desk studies are however required to establish the requirements for design, intensity and sampling to deliver the required accuracy of estimates from any such surveys.

DW & AW Workshop recommendations on biological studies

- Collect life history information (growth, maturity, fecundity etc.); coordinate between key agencies;
- Tag recapture studies to determine habitat utilization and movement.
- Identify additional past and present spawning aggregation sites and characterize migration corridors;
- Define the spatial scale of migrations by individuals participating in spawning aggregations through tag and release studies;
- Conduct studies on temporal variability of oceanographic processes in relation to larval dispersal and connectivity of platforms of currently managed stock units;
- Examine early larval dispersal patterns using genetic markers and otolith microchemistry;
- Investigate population genetic structure of mutton snapper “stocks” within US Caribbean and in relation to the wider Caribbean.
- Examine ontogenetic shifts in habitat usage and diel foraging patterns.

The RP endorses the need for estimates of biological parameters determining productivity (growth, maturity, fecundity). Growth estimates by sex are needed for length-based models, and growth and maturity data are needed for development of biological reference points for exploitation.

The RP endorses the need for better information on distribution and seasonal/ontogenetic migrations and dispersal of mutton snapper. Whilst such information may not necessarily feed directly into stock assessment models, it is important for interpreting CPUE data, evaluating the impact of effort redistribution during closures, and establishing the possibility for over-fishing of

localized populations with limited dispersal and mixing. Modeling of egg and larval drift provides further information on connections between spawning and recruitment sites and the linkages between mutton snapper populations around Puerto Rico and the US Virgin Islands, and is an important long-term area of research rather than for assessing local stock status.

Queen Conch

Data Workshop

Commercial

1. Continuous biological sampling in the Virgin Islands at sufficient levels to adequately characterize size and age composition
2. Link biostatistical data for a fishing trip from Puerto Rico to all of the landings records for that trip.
3. Ensure that the catch and effort data of individual fishers in Puerto Rico can be identified over time.
4. Eliminate the need for expansion factors by obtaining information on all landings.

Recreational

1. Conduct surveys to estimate the magnitude of the U. S. Virgin Islands recreational landings for all species including conch and lobster. It is possible that using a Virgin Islands contractor would improve the likelihood of success of the survey.
2. Include conch and lobster in the MRFSS for Puerto Rico.
3. To adequately characterize catch rates and sizes of mutton snapper caught by recreational anglers in Puerto Rico, very substantial increases in dockside sampling will be needed.

Indices of Abundance

- 1) Fisheries-independent survey efforts currently rarely include stations in deep water, the preferred habitat of adult mutton snapper and adult yellowfin grouper. In addition, large aggregations of queen conch have been reported in deep water by commercial fishers. The group highly recommends the initiation and continued funding of such surveys. As trends can be regional in nature, the group highly recommends that such surveys be conducted throughout Puerto Rico and the US Virgin Islands.
- 2) The commercial landings data from Puerto Rico and the US Virgin Islands have been incompletely entered and a variety of problems are known to exist in those data. The group strongly recommends that every effort be made to resolve the problems with those data. This should include extensive meetings with port samplers and others familiar with the US Caribbean fisheries.

- 3) The group recommends that tag-recapture studies of mutton snapper, yellowfin grouper, and queen conch be conducted in Puerto Rico and the US Virgin Islands to determine habitat utilization and movement of those species.
- 4) Ongoing long-term monitoring studies should be expanded spatially and include data useful for stock assessment, e.g. size-frequency and density information.
- 5) It is suggested that areas exploited by fishermen be compared to those areas where monitoring has been ongoing to further knowledge of essential habitat for these species and improve the design of monitoring efforts (i.e., ensure that monitoring is reflective of fished conditions).
- 6) The group recommends that efforts be made to monitor spawning aggregations of finfish to improve measures of population abundance. Collection of historical indicators of spawner abundance (e.g., directed visual census, analysis of catch statistics for spawning peaks, etc).
- 7) The group encourages the collection and documentation, for this and future Caribbean assessments, of historical information for qualitative and/or quantitative comparisons of current conditions.

Assessment Workshop

1. The efforts to analyze the available data were greatly enhanced by the presence of local fishers and agency representatives. However, there was no local representative from the USVI Division of Fish and Wildlife assigned to the meeting, while the Puerto Rico representative could not attend the full term of the meeting. There must be greater buy-in from the local agencies such that knowledgeable representatives are present for the full term of the meeting. Greater efforts should be made to attract the participation of local fishers.
2. Data from past density surveys should be re-analyzed so that values can be expanded on the basis of both habitat and depth, including confidence limits. Habitats should be matched to those available for existing/planned habitat maps. As a subportion of this, the data for the Puerto Rico 1986 survey should be entered into electronic and GIS formats. This could be done using NOAA's Data Rescue funds.
3. Expansion factors for both Puerto Rico and the USVI should be calculated for conch fishers only.
4. Assessment of the spatial and temporal variations and dynamics of the resource, fishery, habitat and species interactions would be greatly enhanced if traditional ecological knowledge were obtained from fishers. Efforts should be made to incorporate fishers into the process, particularly using NOAA's CRP funds.
5. The impact of the recreational fishery is unknown and must be quantified.

6. Considering the established and potential value of resource surveys, mechanisms should be identified to increase their aerial coverage.

7. More detailed spatial expansions of survey densities should be planned in preparation of the 2010 Conch Update. For this, significant improvements in available data and analyses are required, including but not limited to the following:

- A. Detailed bathymetry data for PR and USVI
- B. Analysis of the impact of closed areas
- C. Inclusion of more detailed habitat maps for the PR western platform currently in progress
- D. Quantified size/age structure of the exploitable stock.

8. The only estimate to date of fishing mortality came from a tagging study in the 1980's. New tagging studies should be initiated to quantify rates of exploitation. This would allow existing SPR models for conch to be used in assessments.

9. Another issue remaining is to investigate the potential impact of very old conch in deep refuges, especially with respect to reproduction, coupled with studies to age very old conch. Such refuges may be substation off St. Thomas/St. John, in patches in Puerto Rico and potentially in protected areas on all three platforms.

10. Intersessional data evaluation workshops for CFMC managed species or species complexes should be conducted by the Council so that SEDAR level analyses are limited to those where data are sufficient to warrant such an analysis.

11. There needs to be a complete review of the potential data collection programs, including commercial and recreational catch, biostatistical sampling and fishery independent surveys for Puerto Rico and US Virgin Islands with the purpose of identifying what relevant information could be obtained and modifying sampling procedures accordingly, including the identification of key economic and ecological indicator species.

Review Workshop

The Review panel replied to Term of reference 9.

- 9) Review the research recommendations provided by the Data and Assessment workshops and make any additional recommendations warranted. Clearly indicate the research and monitoring needs that may appreciably improve the reliability of future assessments. Recommend an appropriate interval for the next assessment.

The RP reviewed the wide range of research recommendations provided by the DW and AW in relation to immediate and longer-term needs for improving the assessment of the stocks and the provision of management advice. The RP provided additional recommendations where appropriate. The research recommendations are reviewed in a separate section of this report (see below under Additional Comments).

It is recommended that the next assessment be deferred until an inter-sessional data evaluation workshop can demonstrate significant progress in the improvement of resource survey and landings data. An interval of 3 years would be appropriate for an inter-sessional workshop.

Additional Comments of the Review Panel

The conclusion that the available data on queen conch fisheries and stock abundance around Puerto Rico and USVI are inadequate to allow a stock assessment or calculation of benchmark statistics was strongly endorsed by the RP. A commitment to long-term research and data collection to address these deficiencies in data and knowledge is essential for effective management supported by robust assessments, and adequate resources need to be provided to collect essential data to support scientifically based management of queen conch in the region. The RP however recognizes the significant effort that has been put into data collection in the region and emphasizes that these have provided a valuable framework for identifying the priorities for future data collection to support stock assessment and fishery management. The DW and AW have made a number of recommendations for future research and monitoring which are reviewed below alongside further recommendations of the RP.

Recommendations of the Data Workshop

Life History

The DW made no recommendations for future research into queen conch life history. Biological parameters for queen conch are generally well characterized from the literature, although variations in growth and maturation over small spatial scales mean that there is uncertainty about area-specific parameters. This is not presently a limiting factor for stock assessment, principally because there are neither data nor model structures available for analytical assessment of conch stocks, but use of yield or spawner per recruit analyses to develop biological reference points would need to account for this fine scale variation. In common with many other species, empirical information is lacking on natural mortality after early life stages, but assumed values and their relationship with age appear to be adequate at present.

The RP made no specific recommendations for high priority research into conch life history parameters, but there was a general view that more information is needed on stock identity and the spatial scale of population processes at each life stage. Genetic studies indicate population connectivity between different areas of the Caribbean, but this does not preclude the existence of stock units that are effectively self-contained at time scales relevant to stock assessment and fishery management. Modeling of conch larvae dispersal by surface currents may shed some light on this issue.

Commercial Statistics

The DW recommended that Puerto Rico conch landings for recent years should be corrected for the change from reporting uncleaned to reporting cleaned meat weights and that this should be done on a port-by-port basis. Landings included in the DW report were not corrected, but approximate corrections at the scale of the entire Puerto Rico fishery were applied in figures

presented in the AW report and during the RW meeting. The RP agreed that it was a high priority to apply such corrections in presenting time-series of conch landings data.

The DW also made the following recommendations regarding the collection of statistics on the commercial fisheries for the three species considered by SEDAR 14:

DW1) Continuous biological sampling in the Virgin Islands at sufficient levels to adequately characterize size and age composition.

DW2) Link biostatistical data for a fishing trip from Puerto Rico to all of the landings records for that trip.

DW3) Ensure that the catch and effort data of individual fishers in Puerto Rico can be identified over time.

DW4) Eliminate the need for expansion factors by obtaining information on all landings.

The first recommendation relates mostly to finfish, but it is also true that future assessments may benefit from more information on the composition of conch catches. Ideally, continuous sampling should be maintained at sufficient levels to allow calculation of required indicator statistics, but occasional intensive sampling may provide a viable alternative. The definitions of ‘sufficient’ and ‘occasional’ can only be judged in a risk assessment context, the relevant question being what precision around indicator statistics is required for management purposes. In the absence of information on which indicator statistics might be desirable the RP is unable to provide more specific recommendations on sampling of conch catch composition.

Recommendations DW2-4 reflect the urgent need for accurate gear-specific total landings and effort data across the whole of the assessed area, and for comprehensive qualifying data to be matched with individual catch records such that meaningful and properly standardized CPUE estimates can be calculated. The RP regards these recommendations as being of the highest possible priority. Future progress in developing stock assessments and population benchmarks for queen conch depends critically on the availability of comprehensive, quantitative information on fishery removals and the associated fishing effort. This will remain true even if, as seems likely, fishery independent indices are used as the primary source of information on stock abundance. ABCs and related statistics will always need to be calculated with reference to complete landings data. The RP further recommends investigation of uncertainty around estimated expansion factors and hence around estimated total landings. This might be achieved by bootstrap sampling of the reported landings data, preferably on a species-specific basis. There also needs to be some evaluation of the assumption that available landings declarations are representative of all license holders. One possible approach would be to examine fishery returns from long-term license holders who have only recently submitted logbook records. If this subset of records is representative of the whole it would be reasonable to suppose that the calculated expansion factors are not biased.

The RP also recommends exploration of alternative approaches to estimating total landings and fishing effort directed at queen conch. These might include randomized sampling of catches at

landing sites, aimed at statistical estimation of landings quantities that might circumvent the possible biases involved in expanding incomplete log-book records. Another approach that could be considered is the use of internet forms to allow fishers to enter catch and effort data directly. In this context it is worth emphasizing the desirability of developing partnerships with local fishers to collect data and to conduct research.

Recreational Fishing

In common with other species, the recreational catch of queen conch may be considerable. Recreational fish catch estimates for Puerto Rico are available for 2000 onwards, but unfortunately conch and other invertebrates were not included in the MRFSS. Based on a one-off survey in 2000, recreational catches of conch were estimated to be at a level of about a third of the commercial landings by Puerto Rico in 2000-20011, i.e. around a quarter of the total landings. Clearly, the recreational catch of queen conch is an important omission from the Puerto Rico total landings data for other years and from the USVI total landings data in all years2. Furthermore, it is difficult to interpret even relative trends without more information on the variability of recreational effort between years3. The DW made the following recommendations relevant to recreational fishing for queen conch:

DW5) Conduct surveys to estimate the magnitude of the U. S. Virgin Islands recreational landings for all species including conch and lobster. It is possible that using a Virgin Islands contractor would improve the likelihood of success of the survey.

DW6) Include conch and lobster in the MRFSS for Puerto Rico.

The RP strongly endorses these two recommendations for both Puerto Rico and USVI and considers that they should be given high priority in the immediate term. Information on total landings is crucial for calculation of ABCs and associated benchmark statistics. The RP further recommends that, in common with the expansion factors for commercial landings statistics (see above), the uncertainty around the current and future recreational landings estimates be investigated. The current figures for Puerto Rico can be regarded as indicative rather than definitive estimates, and the application of the same expansion factor to USVI is somewhat tenuous. Unlike the commercial landings, it would be unrealistic to suppose that 100% coverage of recreational landings could ever be achieved. This makes it important to characterize the uncertainty around all recreational landings estimates.

1 The figures appear to be derived from the observation that recreational catches during a 3 month period were at around 50% of the reported landings over the same period. No adjustment seems to have been made for differences in commercial reporting rates between years.

2 Tentative estimates for the USVI have been made for the same years, assuming that the same relationship exists with commercial landings (SFA Amendment, 2005).

3 The AW report also mentions a similar proportion of recreational landings (35% of commercial landings) for Puerto Rico in 1986, but this would be a slender basis from which to infer a constant proportional contribution over time.

Indices of Abundance

Both fishery-dependent and fishery-independent indices were examined by the DW. A number of recommendations were made on the analysis of CPUE, mainly concerning filtering of trip records and adjustment for reporting cleaned or uncleaned meat weights.

These recommendations were taken on board by the AW, but owing to the lack of relationship between conch density and the ability of commercial divers to catch their daily quotas the resulting indices were considered not to be informative of stock abundance. The RP agrees that under current fishing practices it is unlikely to be feasible to measure diver effort in any way that would allow calculation of CPUE values that are responsive to abundance changes. The RP considers that low priority should be given to further analyses of queen conch CPUE data, given the likelihood that fishery-independent stock indices will be used as the main source of information on stock status in the near future, but the situation might change if alternative effort measures could be devised and recorded. This does not, of course, mean that reduced emphasis should be placed on collecting reliable records on fishing effort. Examination of effort trends is an important component of monitoring for overall fishery 'health', and trends in effort directed at queen conch may in themselves be indicative of changes in abundance.

The DW provided additional recommendations on indices of abundance for species considered by SEDAR 14, of which the following are relevant to queen conch:

DW7) Fisheries-independent survey efforts currently rarely include stations in deep water, the preferred habitat of adult mutton snapper and adult yellowfin grouper. In addition, large aggregations of queen conch have been reported in deep water by commercial fishers. The group highly recommends the initiation and continued funding of such surveys. As trends can be regional in nature, the group highly recommends that such surveys be conducted throughout Puerto Rico and the US Virgin Islands.

DW8) The commercial landings data from Puerto Rico and the US Virgin Islands have been incompletely entered and a variety of problems are known to exist in those data. The group strongly recommends that every effort be made to resolve the problems with those data. This should include extensive meetings with port samplers and others familiar with the US Caribbean fisheries

DW9) The group recommends that tag-recapture studies of mutton snapper, yellowfin grouper, and queen conch be conducted in Puerto Rico and the US Virgin Islands to determine habitat utilization and movement of those species.

DW10) Ongoing long-term monitoring studies should be expanded spatially and include data useful for stock assessment, e.g. size-frequency and density information.

DW11) It is suggested that areas exploited by fishermen be compared to those areas where monitoring has been ongoing to further knowledge of essential habitat for these species and improve the design of monitoring efforts (i.e., ensure that monitoring is reflective of fished conditions).

DW12) The group encourages the collection and documentation, for this and future Caribbean assessments, of historical information for qualitative and/or quantitative comparisons of current conditions.

The RP agreed that all of these recommendations are valid for queen conch, with varying degrees of urgency, but more specific information is required on precisely what is needed and on the proposed methods of addressing them. Recommendation DW8 regarding commercial landings data should be clarified. The high priority that should be given to attempts to improve compliance with reporting requirements has already been noted above. However, the completeness of commercial fishing records is less of an issue in the context of abundance indices, principally because commercial CPUE is not informative as an index. It is nonetheless desirable to ensure that qualifying data for landings records are as comprehensive as possible, for example allowing the efforts of individual fishers to be followed. Significant progress with identifying improved measures of effort may change the priority of this recommendation.

Recommendations DW7, DW10 and DW11 are relevant to fishery-independent surveys, specifically visual surveys that generate habitat-specific queen conch density estimates that can be expanded to domain-wide stock abundance estimates. This assessment method is the most promising for queen conch stock assessments in the near future; the RP took the view that high priority should be given to expanding the spatial coverage and habitat coverage of the fishery-independent surveys with a view to improving the precision of stock abundance estimates. This type of assessment might also benefit from the inclusion of information on how fishing effort is distributed between areas of similar habitat, so that, for example, fishing intensity as well as habitat classification could be used to stratify the density estimates.

The RP agreed that tagging studies of queen conch should be conducted in both Puerto Rico and USVI. Recommendation DW9 relates to the use of tagging to determine patterns of movement and habitat utilization. The RP endorses this recommendation as a medium- to long-term priority, and further recommends that serious consideration should be given to tagging sufficient numbers of conch to allow conclusions to be drawn about population dynamics as well as movement patterns. Modeling of recaptures potentially allows estimation of, among other parameters, rates of both fishing and natural mortality. Even if large-scale, long-term tagging studies prove not to be feasible, short-term, intensive tagging experiments conducted alongside the fishery can be extremely informative, particularly if both commercial and experimental operations are used to generate recaptures and recoveries.

Recommendation DW12 relates to the collation of conch density estimates from different areas of the Caribbean experiencing varying levels of exploitation. Preliminary results of such an exercise constructed by the AW show considerable promise as an innovative approach to placing survey findings in the context of potential population benchmarks. The RP considered that progress with this approach is a high, short-term priority, and recommended that further attention be paid to the influence of habitat type and stock structure (juveniles and adults) on the comparisons. The RP also took the view that the establishment of Marine Protected Areas in the waters of Puerto Rico and USVI holds potential for shedding light on unfished conch densities in the area. This might provide an improved basis for calculating a Bmsy proxy than comparisons with quasi-unexploited densities in other areas of the Caribbean. The AW suggested that conch

densities measured around Puerto Rico are below the ‘Allee effect limit’, this being the threshold below which reduced reproductive output may be expected based on studies in the Bahamas. The RP recommends examination of whether the mean conch densities reported are representative of effective local densities that may exist in patches, hence whether Puerto Rico conch stocks are in fact reproductively compromised to the extent shown. This would need to be addressed before precautionary advice could be offered on the basis of such evidence.

Recommendations of the Assessment Workshop

The AW rejected the use of production (biomass dynamic) models for assessing Puerto Rico queen conch stocks, on the grounds that landings data are incomplete, lacking particularly the recreational component, and that CPUE data do not effectively index stock abundance. Similarly, for reasons stated above, diver CPUE data alone cannot be used to infer trends in stock abundance. The RP agreed with these conclusions and with the decision of the AW to concentrate primarily on fishery-independent surveys. The RP further recommended that stock assessments based on primarily on fishery-dependent data should not be attempted until it can be demonstrated that landings data are complete and that there are informative indices of stock abundance.

The AW compiled the following list of research recommendations for queen conch:

AW1) The efforts to analyze the available data were greatly enhanced by the presence of local fishers and agency representatives. However, there was no local representative from the USVI Division of Fish and Wildlife assigned to the meeting, while the Puerto Rico representative could not attend the full term of the meeting. There must be greater buy-in from the local agencies such that knowledgeable representatives are present for the full term of the meeting. Greater efforts should be made to attract the participation of local fishers.

AW2) Data from past density surveys should be re-analyzed so that values can be expanded on the basis of both habitat and depth, including confidence limits. Habitats should be matched to those available for existing/planned habitat maps. As a sub-portion of this, the data for the Puerto Rico 1986 survey should be entered into electronic and GIS formats. This could be done using NOAA’s Data Rescue funds.

AW3) Expansion factors for both Puerto Rico and the USVI should be calculated for conch fishers only.

AW4) Assessment of the spatial and temporal variations and dynamics of the resource, fishery, habitat and species interactions would be greatly enhanced if traditional ecological knowledge were obtained from fishers. Efforts should be made to incorporate fishers into the process, particularly using NOAA’s CRP funds.

AW5) The impact of the recreational fishery is unknown and must be quantified.

AW6) Considering the established and potential value of resource surveys, mechanisms should be identified to increase their aerial coverage.

AW7) More detailed spatial expansions of survey densities should be planned in preparation of the 2010 Conch Update. For this, significant improvements in available data and analyses are required, including but not limited to the following:

A Detailed bathymetry data for PR and USVI

B Analysis of the impact of closed areas

C Inclusion of more detailed habitat maps for the PR western platform currently in progress

D Quantified size/age structure of the exploitable stock.

AW8) The only estimate to date of fishing mortality came from a tagging study in the 1980s. New tagging studies should be initiated to quantify rates of exploitation. This would allow existing SPR models for conch to be used in assessments.

AW9) Another issue remaining is to investigate the potential impact of very old conch in deep refuges, especially with respect to reproduction, coupled with studies to age very old conch. Such refuges may be substation off St. Thomas/St. John, in patches in Puerto Rico and potentially in protected areas on all three platforms.

AW10) Inter-sessional data evaluation workshops for CFMC managed species or species complexes should be conducted by the Council so that SEDAR level analyses are limited to those where data are sufficient to warrant such an analysis.

AW11) There needs to be a complete review of the potential data collection programs, including commercial and recreational catch, bio-statistical sampling and fishery independent surveys for Puerto Rico and US Virgin Islands with the purpose of identifying what relevant information could be obtained and modifying sampling procedures accordingly, including the identification of key economic and ecological indicator species.

The RP was supportive of all recommendations in this comprehensive list, several of which reiterate suggestions by the DW. The RP draws particular attention to recommendations AW2, AW6 and AW7 which provide specific comments on improving and extending the existing fishery-independent surveys and their analyses. Recommendations AW3, AW5 and AW11 relate to improved collection of commercial and recreational fishery statistics, the importance of which has already been emphasized above. The DW suggested tagging studies to examine patterns of movement and habitat utilization; recommendation AW8 suggests extending tagging studies to examine exploitation rates. The RP endorses this recommendation as a priority for the medium-to long-term, with the suggestion that the feasibility of small-scale intensive tagging experiments be examined in addition to more extensive experiments.

Recommendation AW9 is for investigation of the reproductive contribution of very old conch in deep water refuges. Given the implications for spatial management of the resource, and the context this would supply for interpretation of assessment outcomes in relation to potential population benchmarks, this recommendation should be prioritized for the medium term.

The AW made two recommendations (AW1 and AW10) relevant to future queen conch stock assessment meetings. The RP notes recommendation AW10 to conduct inter-sessional data

evaluation workshops. Given the current lack of a definitive stock assessment for conch the RP considers data evaluation workshops to be a high priority and recommends that the next workshop be held within the next 3 years to maintain impetus particularly on improvements to fishery monitoring and resource surveys. The time-scale for future stock assessments would be dictated by the progress demonstrated at these inter-sessional workshops.